

VELOCITY-BASED TRAINING INTERVENTION: EFFECTS ON LOWER LIMB PERFORMANCE IN FEMALE FLOORBALL ATHLETES

Basil Achermann^{1,2*}, Naire Regazzi², Rahel Heynen³, Julia Suter¹, Silvio Lorenzetti^{1,2}

¹ Institute for Data Analysis and Process Design, ZHAW, Switzerland

² Institute for Biomechanics, ETH Zurich, Switzerland

³ Department for Elite Sport, Swiss Federal Institute of Sport Magglingen, Switzerland

*Correspondence: acbasil@ethz.ch

This study investigated the effects of a six-week velocity-based training (VBT) program in female floorball players. The intervention, focused on back squats and trap bar deadlifts. Key performance metrics included movement velocity of RT exercises, sprint speed, jump performance, stop-and-go (SAG) performance, and load-velocity profiles (LVP). Seventeen participants completed 12 training sessions divided into strength (Sessions 1–6) and power (Sessions 7–12) blocks. Significant improvements were observed in VBT metrics, with squat estimated 1RM increasing by 15.3% (from 1.60 to 1.85) and deadlift estimated 1RM by 5.9% (from 1.76 to 1.87), while mean concentric velocity rose by 7.5% in the squat (0.88 to 0.95 m/s) and by 7.3% in the deadlift (0.86 to 0.92 m/s). This study provides valuable insights into tailoring VBT for female athletes in high-demand team sports like floorball.

KEYWORDS: Resistance Training, team sports, back squat, trap-bar deadlift

INTRODUCTION: In team sports, effective physical preparation is critical for optimizing performance and preventing injuries. Resistance training plays a key role in enhancing athletic abilities and complements sport-specific skill development (Makaruk et al., 2024). However, integrating resistance training into the demanding schedules of team sport athletes poses a challenge, as coaches must balance physical conditioning, recovery, and skill development to maximize benefits while avoiding overtraining.

Floorball, known for its high intensity and dynamic demands, underscores the importance of efficient training methods. The sport requires rapid directional changes, sprints, and frequent stopping and starting, placing substantial demands on lower limb strength and neuromuscular control. This need is amplified by the high incidence of ankle and knee injuries among floorball players, with female athletes particularly vulnerable to such injuries due to their increased susceptibility to anterior cruciate ligament lesions (Zech et al., 2022). Enhancing lower limb strength is therefore crucial for improving performance and reducing injury risks in this population. Velocity-based training (VBT) has emerged as a versatile and effective method of resistance training. By using the speed of weight movements to guide the prescription of loads, sets, and repetitions, VBT dynamically adjusts training loads based on an athlete's performance. Notable features of VBT include the ability to estimate the one-repetition maximum (1RM) from submaximal efforts, reducing the time and risks associated with frequent maximal strength testing, especially for less experienced athletes and large groups, and the use of velocity zones to target specific strength outcomes (Weakley et al., 2021).

While VBT has demonstrated benefits in improving maximal strength, jump performance, and sprint speed, much of this research has predominantly focused on males. Consequently, there is a notable gap in understanding VBT's effectiveness for females, particularly in team sports like floorball. The present study investigates the effects of a six-week VBT program on lower limb performance in female floorball players. We hypothesise that twice-weekly 30-min VBT sessions over six weeks will significantly enhance lower-limb strength (estimated 1RM, load-velocity profiles (LVP)), and sport-specific performance (sprint speed, jump height, SAG time,) in female floorball players. By targeting these metrics, this study aims to provide valuable insights into the application of VBT for female athletes in team sports

METHODS: This study was designed as a longitudinal experimental intervention to investigate the effects of VBT on lower limb performance. The intervention consisted of 12 training

sessions over 6 weeks, with assessments conducted before (T1) and after (T2) the intervention.

Participants: A total of 23 female participants were initially enrolled in the study, of which 17 completed the intervention ($n = 17$). Participant characteristics at baseline were as follows: body weight = 64.5 ± 9.1 kg, height = 1.69 ± 0.06 m, and age = 22.0 ± 3.7 years. Participants were recruited from a local floorball team and had to meet specific inclusion criteria, including active membership in the team, being over 18 years old and having previous experience in strength training. Exclusion criteria were any recent injuries that could affect participation in training or testing procedures. All participants provided written informed consent before participating in the study, in accordance with ethical research guidelines.

Testing: Performance assessments were conducted at T1 and T2 to evaluate key lower limb performance metrics. These assessments included sprint performance, stop-and-go (SAG) ability, jump performance through countermovement jump (CMJ) and squat jump (SJ), and LVPs. The pre-intervention LVP was also used to determine individualized starting loads, ensuring an appropriate training intensity for each participant. Before testing, participants completed a self-selected warm-up routine, which included aerobic exercises, joint mobilization, and dynamic movements, similar to their regular floorball training preparation. Participants carried out at least two maximal 20 m sprints, with times recorded at 5 m and 20 m using photocell timing gates (Witty Timing System, Microgate, Bolzano, Italy) positioned at 0, 5, and 20 m. Participants performed CMJ and SJ on a force platform (CYCESS SP Sport, Innsbruck, Austria). Prior to testing, participants completed a standardized warm-up consisting of four CMJs, four SJs, four consecutive jumps with knees to chest, and six ankle jumps. For each jump type, three valid attempts were recorded, with the best attempt retained for analysis. An incremental loading test was conducted for the back squat and trap bar deadlift exercises. For the back squat, participants started from an upright position, descended until their thighs were parallel to the floor (approximately 90° knee angle), and then executed a maximal effort concentric movement to return to the start position. The velocity of the concentric phase was recorded, with the mean concentric velocity (MCV) as the primary performance variable. The load increment protocol included predetermined weight steps of 15 kg (3 reps), 30 kg (3 reps), 40 kg (3 reps), 50 kg (2 reps), and 60 kg (2 reps). For the trap bar deadlift, participants stood centered within the bar, feet shoulder-width apart, gripping the handles with palms facing inward. They performed a maximal concentric lift to a fully upright position and then controlled the descent. The load increment protocol included fixed weight steps starting at 35 kg (3 reps), followed by 45 kg (3 reps), 55 kg (3 reps), 65 kg (2 reps), and 75 kg (2 reps). The load steps for both exercises at T1 were replicated at T2 to ensure consistency in the testing protocol and allow pre/post comparison.

Intervention: The training intervention consisted of a 6-week VBT program, with participants completing two sessions per week, totaling 12 planned sessions. To account for any missed sessions, two reserve sessions were included, although the primary analysis focused on the 12 planned sessions. The intervention was divided into two distinct blocks to target specific force-velocity profile zones. The first block (Sessions 1–6) emphasized strength development, with training loads adjusted to maintain the MCV within a range of 0.55–0.7 m/s. The second block (Sessions 7–12) shifted the focus to power development, with a target MCV range of 0.8–1.0 m/s [17]. Velocity was the primary target for load adjustments: if three or more repetitions in a strength set, or five or more in a power set, fell outside the target velocity range, the load for the next set was adjusted by ± 5 kg. Movement velocity was continuously monitored using the GymAware PowerTool, providing real-time feedback to ensure that the training loads aligned with the target velocity zones for each block.

Statistical Analysis: Data are presented as mean $\pm$ standard deviations (SD) and processed using the statistical software R Studio (Version 4.4.1). The Shapiro-Wilk test was used to assess the normality of the data distribution. Paired t-tests were utilized for pre/post

comparisons where data met normality requirements, while the Wilcoxon signed-rank test was employed as a non-parametric alternative when normality was not met. Effect sizes (ES) were calculated using Cohen's *d* to quantify the magnitude of changes observed. The significance level was set at $p \leq 0.05$. All *p*-values were corrected for multiple testing using the false discovery rate (FDR). Effect sizes (ES) were classified according to Cohen 150 [23] as small (≥ 0.2), medium (≥ 0.5), or large (≥ 0.8).

RESULTS: The results of the tests conducted before and after the training intervention are summarized in Table 1. In addition, a notable increase in body weight was observed over the intervention period, with a mean change of 2.15% ($\Delta = 1.39$ kg, $p = 0.001$)

Table 1: Results of pre and post testing.

	T1	T2	$\Delta / \Delta\%$	p-values	ES
Sprint 5m (s)	1.15 $\pm$ 0.1	1.16 $\pm$ 0.1	0.01/1.2%	0.593	small
Sprint 20m (s)	3.41 $\pm$ 0.1	3.40 $\pm$ 0.1	-0.01/-0.3%	0.13	small
SAG (s)	5.01 $\pm$ 0.2	4.97 $\pm$ 0.2	-0.04/-0.7%	0.025	small
CMJ Height (m)	30.2 $\pm$ 4.1	30.3 $\pm$ 3.3	0.01/0.6%	0.464	small
CMJ rel. Power (W/kg)	46.8 $\pm$ 5.1	47.7 $\pm$ 5.8	0.9/2.1%	0.048	small
SJ Height (m)	28.3 $\pm$ 4.2	29.4 $\pm$ 3.5	1.1/3.9%	0.027	small
SJ rel. Power (W/kg)	44.4 $\pm$ 5.6	46.5 $\pm$ 5.4	2.1/4.8%	< 0.001	large
Squat AV (m/s)	0.88 $\pm$ 0.1	0.95 $\pm$ 0.1	0.07/7.5%	0.016	medium
Squat est. 1RM (ratio)	1.60 $\pm$ 0.3	1.85 $\pm$ 0.4	0.25/15.3%	0.003	medium
Deadlift AV (m/s)	0.86 $\pm$ 0.1	0.92 $\pm$ 0.1	0.06/7.3%	0.002	large
Deadlift est. 1RM (ratio)	1.76 $\pm$ 0.3	1.87 $\pm$ 0.2	0.10/5.9%	0.006	medium

Changes in athletic performance variables and load-velocity profiles from T1 to T2. Data are mean $\pm$ SD; T1: Pre intervention testing; T2: Post intervention testing; Δ : Absolute change; $\Delta\%$: Percentage change; *: *p* value Statistical significance; ES: Effect size (Cohen's *d*); SAG: Stop-and-Go; CMJ: Countermovement jump; SJ: Squat jump; AV: Average MCV against all absolute loads; est. 1RM: Estimated one-repetition maximum relative to BW.

Statistical improvements were observed in performance metrics for both the squat and trap bar deadlift performance variables from T1 to T2. For the average velocity (AV), consistent increases were noted across both exercises. The squat improved by 7.5% ($p = 0.016$), while the trap bar deadlift increased by 7.3% ($p = 0.002$). In terms of the estimated one-repetition maximum relative to body weight (1RMest), significant gains were observed for both exercises. The squat 1RMest increased by 15.3% ($p = 0.003$), and the trap bar deadlift 1RMest rose by 5.9% ($p = 0.024$). The SJ results demonstrated significant increases in maximum relative power by 4.8% ($\Delta = 2.1$ W/kg, $p < 0.001$).

DISCUSSION: This study investigated the impact of a 6-week VBT program, consisting of brief 30-minute sessions, on lower limb performance in female floorball athletes. The findings revealed notable improvements in strength and power, although gains in specific athletic tasks such as jumping, sprinting, and stop-and-go ability were limited. These mixed results reflect the strengths and limitations of the intervention design and offer valuable insights for optimizing future training protocols. The VBT program led to clear improvements in the RT domain, as demonstrated by enhanced load-velocity profiles and increases in estimated 1RM. The AV, showed a measurable improvement from pre- to post-testing, indicating more efficient force production and improved power output. These changes are consistent with prior research highlighting the role of VBT in maximizing athletic performance (Rauch et al., 2018). However, task-specific performance outcomes, such as jump height, sprint ability, and stop-and-go performance, did not improve significantly. While the relative power of the SJ improved

significantly, CMJ relative power gains were less pronounced, and jump height improvements were not significant for either jump type. This outcome aligns with findings by (Rauch et al., 2018), who observed similar results using a comparable VBT protocol. However, given that several studies using diverse VBT programs have demonstrated improvements in jump performance, it is somewhat surprising that this study did not observe similar improvements. This discrepancy underscores the complexity of jump performance adaptations and suggests that further exploration of training parameters, such as load selection and exercise specificity, may be necessary. Interestingly, SJ metrics, including relative power and jump height, showed better outcomes compared to CMJ. This finding contrasts with the typical pattern, where CMJ performance is usually superior to SJ performance. The discrepancy observed in this study may reflect the closer similarity of the squat jump to the primary exercises included in the VBT program, such as the squat and trap bar deadlift. Both the SJ and these exercises emphasize concentric force production, which was the focus of the VBT protocol. These similarities may explain the more pronounced improvements in SJ metrics compared to CMJ performance.

Limitations: The small sample size ($n = 17$) and lack of a control group limit generalisability and preclude definitive causal inferences, while the six-week duration may have been too short to elicit meaningful sprint and SAG adaptations. Furthermore, variability in squat technique and range of motion among participants may have limited the transfer of strength and power gains to performance. Additionally, reliance on estimated 1RM (rather than true 1RM testing) introduces estimation error, and the relatively under-resourced training environment for female floorball players may have constrained their physical development compared to athletes in better-supported sports.

CONCLUSION: This study demonstrated that a time-efficient, 30-minute VBT program consisting of back squats and trap bar deadlifts effectively enhanced RT performance in the lower limbs of female floorball athletes. The intervention led to measurable improvements in strength, and power, highlighting the potential of VBT as a practical and impactful training modality that can be integrated within demanding athletic schedules. These results underscore the feasibility of incorporating brief resistance training sessions alongside regular team practices. However, these gains did not fully translate to sport-specific performance outcomes such as sprint, jump, and stop-and-go ability. Future research could investigate the refinement of VBT methods, such as optimizing velocity zones to enhance individualization and fatigue management. Extending training durations and integrating sport-specific components, such as plyometric exercises, may enhance the translation of RT adaptations to sport-specific tasks. Such advancements could guide the development of more effective and comprehensive training programs for female athletes in team sports.

REFERENCES

- Makaruk, H., Starzak, M., Tarkowski, P., Sadowski, J., & Winchester, J. (2024). The Effects of Resistance Training on Sport-Specific Performance of Elite Athletes: A Systematic Review with Meta-Analysis. *Journal of Human Kinetics*, 91, 135–155. <https://doi.org/10.5114/jhk/185877>
- Rauch, J. T., Loturco, I., Cheesman, N., Thiel, J., Alvarez, M., Miller, N., Carpenter, N., Barakat, C., Velasquez, G., Stanjones, A., Aube, D., Andersen, J. C., & De Souza, E. O. (2018). Similar Strength and Power Adaptations between Two Different Velocity-Based Training Regimens in Collegiate Female Volleyball Players. *Sports*, 6(4), 163. <https://doi.org/10.3390/sports6040163>
- Weakley, J., Mann, B., Banyard, H., McLaren, S., Scott, T., & Garcia-Ramos, A. (2021). Velocity-Based Training: From Theory to Application. *Strength & Conditioning Journal*, 43(2), 31–49. <https://doi.org/10.1519/SSC.0000000000000560>
- Zech, A., Hollander, K., Junge, A., Steib, S., Groll, A., Heiner, J., Nowak, F., Pfeiffer, D., & Rahlf, A. L. (2022). Sex differences in injury rates in team-sport athletes: A systematic review and meta-regression analysis. *Journal of Sport and Health Science*, 11(1), 104–114. <https://doi.org/10.1016/j.jshs.2021.04.003>

ACKNOWLEDGEMENTS: The author would like to thank all the participants who made this work possible.